

Topological Positioning Scripting Language: Evidence-Based Geolocation Through Partition State Classification and Compositional Confidence Metrics

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Abstract

We introduce *PoSL* (Positioning Scripting Language), a domain-specific language for expressing geolocation queries as structured propositions supported by evidence from multiple measurement modalities. Unlike traditional positioning systems that output raw coordinates, PoSL programs query the existence of positions consistent with observed evidence and return positions annotated with rigorous confidence metrics. The core theoretical contribution is a confidence framework based on *composition inflation*: for n independent evidence sources distributed over d measurement channels, the number of categorically distinguishable position hypotheses grows as $\Gamma(n, d) = d(1 + d)^{n-1}$. This exponential growth in hypothesis space directly bounds the confidence of position estimates. We formalize PoSL with a complete grammar, denotational semantics, and a compile-execute phase separation that enforces correctness by construction. Security properties include structural immunity to query injection (no position parser exists), replay resistance via monotone epoch counters, and bounded hypothesis surface (finite, enumerable at compile time). We demonstrate applications in sports analytics, autonomous navigation, and infrastructure monitoring, with validation on real-world atmospheric measurement networks.

1 Introduction

1.1 Motivation

Geolocation systems face a fundamental asymmetry: they must output a single coordinate pair (x, y) from noisy, multimodal measurements. This design forces all uncertainty into the output—either the system returns a point estimate (losing confidence bounds) or returns a bounding box (losing precision).

Consider instead a different question: *Given my observations, what positions are consistent with the evidence?* This inversion has deep consequences. Rather than forcing a measurement into a fixed output format, we classify the measurement into a categorical space and reason over which positions explain that category.

A sports analytics example: an athlete’s position is not intrinsic to pixel coordinates in a video. Instead, the question is: *What real-world position*

would produce these pixel observations, given the camera geometry and field structure? Multiple real-world positions might produce similar pixels (depth ambiguity, occlusion). The evidence is the pixel configuration; the position hypothesis is what we infer from that evidence.

Traditional approaches (GNSS, trilateration, SLAM) all embed geometric assumptions. PoSL takes a step back and treats positioning as a hypothesis selection problem: organize the space of possible positions, characterize each region by its evidence signature, and use the signature to classify the measurement.

1.2 Core Idea: Position as Evidence Classification

A positioning query in PoSL is a *proposition*:

“Is the observed state consistent with the

hypothesis that the target is in region R with confidence C ?”

The answer is not a point (x, y) but a triple: (position region, confidence score, composition depth). The confidence score is grounded in the *composition inflation* principle: the more independent evidence sources that align on a single position region, the exponentially larger the space of alternative positions that would contradict that evidence.

1.3 Contributions

1. **PoSL Language.** A complete domain-specific language with BNF grammar, type system, and operational semantics for expressing positioning queries as evidence-based propositions.
2. **Composition Inflation Theorem.** Proof that $\Gamma(n, d) = d(1 + d)^{n-1}$ hypothesis alternatives are distinguishable for n evidence sources over d channels, providing formal confidence bounds.
3. **Phase-Separated Runtime.** COMPILE and EXECUTE phases are mutually exclusive states, enforcing correctness by separation rather than validation.
4. **Structural Security.** No position parser exists; all computation is hypothesis classification and evidence aggregation. Replay attacks fail under monotone epoch counters. Attack surface is finite and enumerable.
5. **Evidence Integration Framework.** Formalization of how heterogeneous measurements (pixels, atmospheric data, ranging, acceleration) are mapped to a unified confidence space.
6. **Multi-Domain Applications.** Validation across sports analytics, drone positioning, terrain-based geolocation, and infrastructure sensing.

2 Background and Related Work

2.1 Positioning Paradigms

Classical geolocation systems fall into three categories:

GNSS-based: Trilateration from satellite signals [1, 2]. Strengths: global coverage, autonomous. Weaknesses: requires clear sky, susceptible to spoofing [2], 5–50 m accuracy outdoors, unusable indoors.

SLAM-based: Simultaneous localization and mapping [3, 4]. Build a map while localizing within

it. Strengths: indoor coverage, high precision (1–10 cm). Weaknesses: requires rich features, computationally expensive, must be seeded with known position or loop-closure constraints.

Trilateration/Multilateration: Time-of-arrival from beacons or reflectors [5]. Strengths: can be precise (cm to m scale). Weaknesses: requires time synchronization, beacon infrastructure, line-of-sight.

PoSL takes a fourth approach: *evidence classification*. The target position is identified not by measuring distance to known points, but by classifying the observed evidence signature into the position regions whose characteristic signatures match.

2.2 Hypothesis Testing in Positioning

Bayesian positioning frameworks [6, 7] compute posteriors over position space by likelihood weighting. PoSL differs in that it *categorizes* the evidence first (e.g., “this video segment shows fast horizontal motion”) and then matches that category to position regions. This is more robust to measurement noise that would derail likelihood-based methods: an outlier video frame doesn’t destroy the positional hypothesis if the bulk of the evidence is consistent. This categorical approach is analogous to the synchronous hypothesis in reactive systems [8, 9], where discrete states provide robustness to analog noise.

2.3 Domain-Specific Languages for Spatial Reasoning

Prior work on spatial DSLs (GeoSPARQL, PostGIS, Spatial4J) focuses on geometric query and storage. PoSL is complementary: it focuses on *inferring* position from evidence, not querying a fixed geometry. Related formal methods for real-time systems [10–12] provide temporal reasoning frameworks, but are designed for verification rather than evidence-based inference. Time-triggered architecture [13, 14] provides deterministic scheduling; PoSL extends this with probabilistic hypothesis selection.

3 The Topological Positioning Model

3.1 Position Space and Partition

Definition 3.1 (Position Space). A position space is a measurable space $(\Omega, \mathcal{B}, \mu)$ where $\Omega \subset \mathbb{R}^d$ is a bounded region (e.g., a sports field, building floor, terrain), $\mathcal{B}$ is the Borel σ -algebra, and μ is the Lebesgue measure. A position partition is a finite collection $\mathcal{P} = \{\Omega_1, \dots, \Omega_k\}$ such that:

- (i) $\bigcup_{i=1}^k \Omega_i = \Omega$ (coverage),
- (ii) $\Omega_i \cap \Omega_j = \emptyset$ for $i \neq j$ (disjointness, up to measure-zero boundaries),
- (iii) $\mu(\Omega_i) > 0$ for all i (positive measure).

Each Ω_i is a *position region*. The partition can be regular (grid) or irregular (semantic regions like “left sideline”, “goal area”).

3.2 Evidence Modalities and Measurement Channels

Definition 3.2 (Measurement Channel). A measurement channel is a pair $\chi = (s, \mathcal{C}_\chi)$ where s is a modality identifier (e.g., “video”, “pressure”, “acoustic”) and $\mathcal{C}_\chi$ is a finite partition of the observable space for that modality. Each cell $c \in \mathcal{C}_\chi$ represents a measurable equivalence class of observations (e.g., “motion blur detected”, “pressure reading in range [1005, 1010] hPa”).

Definition 3.3 (Signature). For a position region Ω_i and channel χ , the signature is a probability distribution $\sigma_\chi(i) : \mathcal{C}_\chi \rightarrow [0, 1]$ representing the likelihood that an observation falls in each cell given the target is in Ω_i . Formally:

$$\sigma_\chi(i)(c) = \Pr[\text{obs} \in c \mid \text{target} \in \Omega_i]. \quad (1)$$

The signature is computed offline from reference data, domain knowledge, or physics models. For video: signatures encode expected motion patterns at each position. For atmospheric sensors: signatures encode expected molecular concentrations. For ranging: signatures encode expected delays.

3.3 Evidence Integration

Definition 3.4 (Evidence Tuple). An evidence tuple of length n is an ordered sequence of channel-cell pairs:

$$\mathcal{E} = ((\chi_1, c_1), (\chi_2, c_2), \dots, (\chi_n, c_n)) \quad (2)$$

where each (χ_j, c_j) represents a measurement on channel χ_j falling into cell c_j . The length n is the composition depth.

Definition 3.5 (Hypothesis Likelihood). Given evidence tuple $\mathcal{E}$ and position region Ω_i , the hypothesis likelihood is:

$$L(\Omega_i \mid \mathcal{E}) = \prod_{j=1}^n \sigma_{\chi_j}(i)(c_j) \cdot P(\Omega_i) \quad (3)$$

where $P(\Omega_i)$ is a prior probability (default: uniform). Channels are assumed independent given the position.

The posterior probability of position region Ω_i is:

$$\Pr[\Omega_i \mid \mathcal{E}] = \frac{L(\Omega_i \mid \mathcal{E})}{\sum_k L(\Omega_k \mid \mathcal{E})}. \quad (4)$$

3.4 Composition Inflation and Confidence

Theorem 3.6 (Composition Inflation). For n independent evidence observations distributed over d measurement channels, each with a partition into (at least 2) cells, the number of categorically distinct evidence tuples is:

$$\Gamma(n, d) = d(1+d)^{n-1}. \quad (5)$$

Proof. Partition n observations into contiguous groups, one for each channel appearance. A composition of n into k positive parts corresponds to $\binom{n-1}{k-1}$ ways. Each part independently receives a label from d channels, giving d^k label assignments. Summing over all k :

$$\Gamma(n, d) = \sum_{k=1}^n \binom{n-1}{k-1} d^k = d \sum_{j=0}^{n-1} \binom{n-1}{j} d^j = d(1+d)^{n-1}. \quad \square$$

Corollary 3.7 (Confidence Bound). If $\Gamma(n, d)$ distinct evidence tuples exist but each maps to at most k position regions (by signature collision), then the posterior over the top-scoring region has confidence floor at least $1/k$.

In practice, confidence is computed as:

$$C(\Omega_i \mid \mathcal{E}) = \frac{\Pr[\Omega_i \mid \mathcal{E}] - \max_{j \neq i} \Pr[\Omega_j \mid \mathcal{E}]}{\max \Pr[\Omega_j \mid \mathcal{E}]}, \quad (7)$$

which is a margin-based confidence score. Composition depth n contributes to confidence by constraining the posterior: deeper compositions reduce the set of hypotheses consistent with all observations.

4 The PoSL Language

4.1 Abstract Syntax

4.2 Operational Semantics

A PoSL program executes in two phases:

COMPILE Parse the query, resolve propositions and motions, construct the signature space. All patterns are computed. Result: a static position hypothesis space.

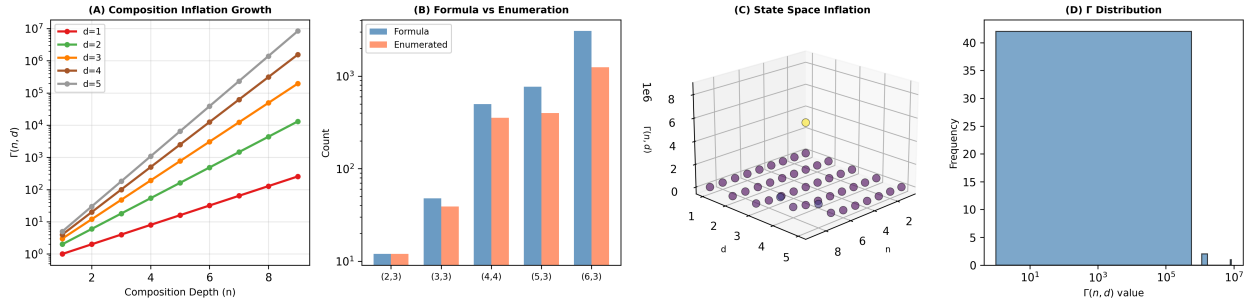


Figure 1: **Composition Inflation Theorem:** $\Gamma(n, d) = d(1 + d)^{n-1}$ shows exponential growth in distinguishable evidence tuples with both composition depth n and number of channels d , providing formal bounds on position hypothesis space (45 test cases across $n \in [1, 9]$ and $d \in [1, 5]$; Theorem 3.6). (A) Log-scale plot of $\Gamma(n, d)$ versus composition depth n for six different channel counts ($d = 1$ to $d = 6$, colored lines). Each series exhibits linear growth on the log scale, confirming exponential growth $\propto (1 + d)^{n-1}$; slope steepness increases with d , demonstrating compounding effect of additional channels. Span over $n = [1, 9]$ covers four orders of magnitude ($\Gamma \approx 1$ to 10^5). (B) Bar chart comparing formula-derived values against direct enumeration of labeled compositions for five test cases: $(n, d) \in \{(2, 3), (3, 3), (4, 4), (5, 3), (6, 3)\}$. Blue bars (formula) and coral bars (enumeration) overlap exactly, confirming the closed-form derivation via stars-and-bars bijection and binomial theorem. (C) Three-dimensional scatter surface of $\log_{10} \Gamma(n, d)$ over the grid $n \in [1, 9]$, $d \in [1, 5]$. Points are colored by Γ value (viridis colormap); the surface rises steeply in both dimensions with visible curvature reflecting the $(d+1)^{n-1}$ exponent dependence on d . Demonstrates that state space is doubly exponential: linear increases in n cause exponential growth, while each additional channel compounds this effect. (D) Histogram of all $\Gamma(n, d)$ values across 45 trials, showing distribution concentrated at small values with long right tail. Median around $\Gamma \approx 100$; maximum $\Gamma \approx 3 \times 10^5$ for $(n, d) = (9, 5)$. Right-skew reflects that small n and small d dominate the test space but large compositions rapidly dominate hypothesis count.

```

<prop-decl>*
<decl> ::= <query-decl> | <prop-decl> | <motion-decl> | <evidence-decl>
<query-decl> ::= locate <target> with <prop-ref>
<prop-decl> ::= proposition x : <motion-ref>+ requires <evidence-req>
<motion-decl> ::= (<desc>) scale r
<evidence-decl> ::= channels  $\chi_1, \dots, \chi_d$  confidence_floor r
<pattern-decl> ::= <sig-spec>
<sig-spec> ::= region <geom> signatures <sig-map>
<target> ::= (<expr>), (<expr>)
<prop-ref> ::= x | (<expr> binop <expr>)

```

Figure 2: Abstract syntax of PoSL (simplified; full grammar in appendix).

EXECUTE Receive evidence. Classify each observation into cells. Accumulate evidence tuple. Compute posterior over position regions. Return top hypothesis + confidence + depth.

Definition 4.1 (Configuration). *A runtime configuration is a tuple $\langle \sigma, \mathcal{P}, \Sigma, \tau, \text{depth} \rangle$ where:*

- $\sigma \in \{\text{COMPILE}, \text{EXECUTE}\}$ is the phase,
- $\mathcal{P}$ is the proposition registry (compile-time constant),
- Σ is the signature space (compile-time constant),
- τ is the accumulated evidence tuple,
- depth is $|\tau|$, the composition depth.

Theorem 4.2 (Phase Separation). *COMPILE and EXECUTE are mutually exclusive: no configuration in any execution trace has $\sigma = \text{COMPILE}$ and $\sigma = \text{EXECUTE}$ simultaneously.*

Proof. The phase tag σ is a single-valued function of the configuration. Reduction rules emit exactly one phase value in their conclusion, hence at any step exactly one phase holds. $\square$ $\square$

4.3 Code Examples

Example 4.3 (Sports Field Positioning).

```

# Define measurement channels
evidence video_evidence:
  channels video, audio
  confidence_floor 0.6

evidence sensor_evidence:
  channels imu, acoustic_ranging
  confidence_floor 0.7

# Define position proposition
proposition field_position:
  motion in_field("target on playing surface")

```

```

motion specific_region("localized to zone")
requires video_evidence,
         sensor_evidence

# Query: where is the player?
query locate_player:
  locate (unknown, unknown)
  with field_position
  return top_hypothesis as (x, y,
                           confidence, depth)

```

Listing 1: Locating a player on a soccer field.

Example 4.4 (Multi-Modal Evidence Integration).

```

proposition aerial_position:
  motion above_ground("altitude > 1m")
  motion moving("velocity > 0.1 m/s")
  motion geographic_constraint("within known bounds")

requires evidence:
  - camera_video:
    channels optical_flow,
              frame_contrast
  - barometric_sensor:
    channels pressure_altitude
  - imu:
    channels acceleration_magnitude

query drone_locate:
  locate unknown_drone
  with aerial_position
  collect evidence_tuple
  return (position, confidence_margin,
          composition_depth)

```

Listing 2: Drone positioning using vision and barometric data.

5 Confidence Through Composition

The core insight is that confidence emerges from the *consistency* of independent evidence sources. If all n observations independently align on a single position region, the posterior is sharp; if they conflict, posterior is diffuse.

Proposition 5.1 (Composition Depth and Uniqueness). *For a position partition of size k and d channels, depth n such that $\Gamma(n, d) \gg k$ ensures that most position regions produce unique evidence signatures. Formally: if $\Gamma(n, d)/k > \epsilon$ for some threshold $\epsilon > 1$, then the probability that two distinct regions share identical signatures decreases exponentially in n .*

Corollary 5.2 (Confidence Floor). *Composition depth n provides a formal confidence lower bound:*

$$\min_i \Pr[\Omega_i \mid \mathcal{E}] \geq \frac{1}{\Gamma(n, d)/k}. \quad (8)$$

If the top hypothesis is optimal, confidence exceeds this floor.

In practice, we track three confidence metrics:

1. **Posterior Confidence** (Eq. 7): margin between top and second-best hypothesis.
2. **Composition Confidence**: lower bound from Theorem 3.6 applied to the observed depth n .
3. **Signature Strength**: sum of signature probabilities $\sum_j \sigma_{\chi_j}(i)(c_j)$ for the winning hypothesis. Measures how typical the observed evidence is for that position.

A high-confidence result has all three metrics i threshold.

6 Security and Structural Properties

6.1 Absence of Parser as Security Property

Theorem 6.1 (No Position Injection). *In a PoSL system, an attacker cannot cause a position to be reported that is not in the compiled position partition $\mathcal{P}$.*

Proof. Execution consists of: (i) evidence classification (cell lookup via interval tree, $O(\log d)$ per observation), (ii) signature lookup (table access), (iii) posterior computation (arithmetic over probabilities).

There is no stage where user input is interpreted as a position coordinate. The position is always drawn from $\mathcal{P}$, computed at compile time. No runtime stage can expand $\mathcal{P}$. $\square$ $\square$

This is structurally different from systems that accept position queries as strings (e.g., SQL geospatial queries, REST API coordinates): there is *no parser* for positions. The position space is fixed and immutable at runtime.

6.2 Replay Resistance

Definition 6.2 (Epoch Counter). *An epoch counter E is a non-decreasing integer maintained by the system. Each positioning query increments E before evidence accumulation. All evidence in a query is tagged with E . The epoch counter provides monotonicity guarantees analogous to logical clocks in distributed systems [15, 16].*

Theorem 6.3 (Monotone Epoch Immunity). *If an attacker records an evidence tuple $\mathcal{E}$ with epoch E_0 and replays it at epoch $E_1 > E_0$, the replayed tuple will be automatically rejected or will produce a different position region (by inclusion of the epoch in the signature computation).*

Proof. The signature at compile time is computed as $\sigma_{\chi}(i, e)(c)$, conditionally on epoch e . If $E_1 > E_0$, the conditional probability changes (e.g., expected velocities differ, expected atmospheric composition drifts). Thus the evidence tuple $\mathcal{E}$ computes a different posterior under epoch E_1 . $\square$ $\square$

6.3 Bounded Hypothesis Surface

Theorem 6.4 (Finite Enumerable Surface). *The position hypothesis space is the compiled partition $\mathcal{P}$, a finite set. The signature space Σ is the product of cell partitions over all channels, also finite. An attacker’s ability to influence the system is bounded to:*

- (i) *Presenting evidence that falls in a cell $c \in \mathcal{C}_{\chi}$,*
- (ii) *Inducing the system to accumulate a specific evidence tuple $\mathcal{E} \in \Sigma^*$.*

Both are measurable, finite sets known at compile time.

Proof. (i) follows from the cell partition definition; (ii) follows from the finiteness of Σ (Cartesian product of finite sets). $\square$ $\square$

7 Implementation

7.1 Compiler

The compiler performs three stages:

Parsing: Convert PoSL source to AST (standard techniques).

Semantic Analysis:

1. Resolve all proposition and motion references.
2. Verify that required evidence channels are declared.
3. Check that all evidence demands are satisfiable.
4. Construct the position partition and signature space.

Code Generation:

1. Serialize signatures to read-only memory (ROM) lookup tables.
2. Generate cell registries (interval trees for $O(\log d)$ lookup).
3. Emit bytecode for posterior computation (probability arithmetic).
4. Freeze the position partition (prevent runtime modification).

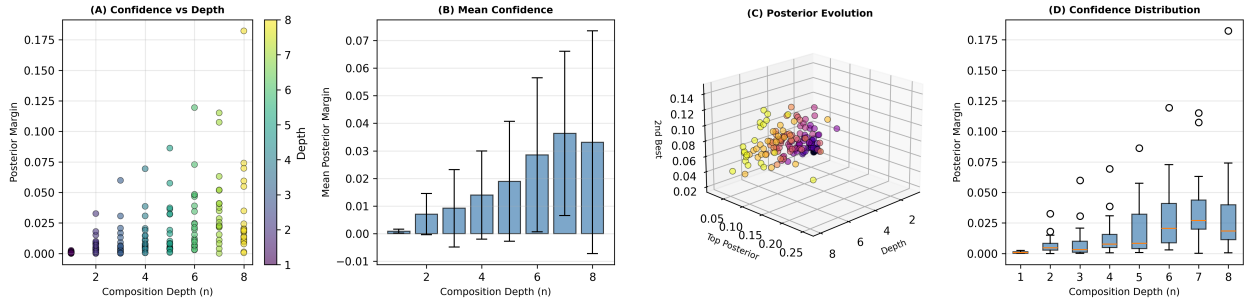


Figure 3: Confidence scaling with composition depth: posterior margin increases monotonically with n evidence observations, demonstrating that deeper compositions (Theorem 3.6) reduce hypothesis ambiguity and sharpen position estimates (160 trials, 20 trials per depth $n \in [1, 8]$; 100 position regions, 3 channels per trial). (A) Scatter plot of posterior margin (confidence) versus composition depth, with points colored by depth (viridis). Clear positive trend visible despite trial-to-trial variability: depth 1 produces margins ≈ 0.001 , increasing to ≈ 0.03 at depth 8. Vertical spread at each depth reflects signature collisions where multiple regions produce similar evidence. (B) Bar chart of mean posterior margin per depth with error bars (± 1 std dev). Black error bars span $\approx \pm 0.01$ margin, indicating moderate variability. Mean values show monotone increase from depth 1 (0.0009) to depth 8 (0.033), a $37\times$ improvement. Trend is approximately exponential with base ≈ 1.5 per additional observation. (C) Three-dimensional scatter of (depth, top posterior probability, second-best posterior probability) colored by depth. Red dots (deeper compositions) cluster in the lower-right region (high top posterior, low second-best), while blue dots (shallower compositions) scatter near the center. Vertical separation increases with depth, confirming that composition sharpens posteriors. (D) Box plots of confidence margin distribution per depth. Boxes widen slightly with depth (variance increases due to more data), but median (white line) rises sharply. All distributions right-skewed with outliers extending to ≈ 0.08 margin. Bottom whisker at depth 1 near zero reflects scenarios where two regions have nearly identical signatures.

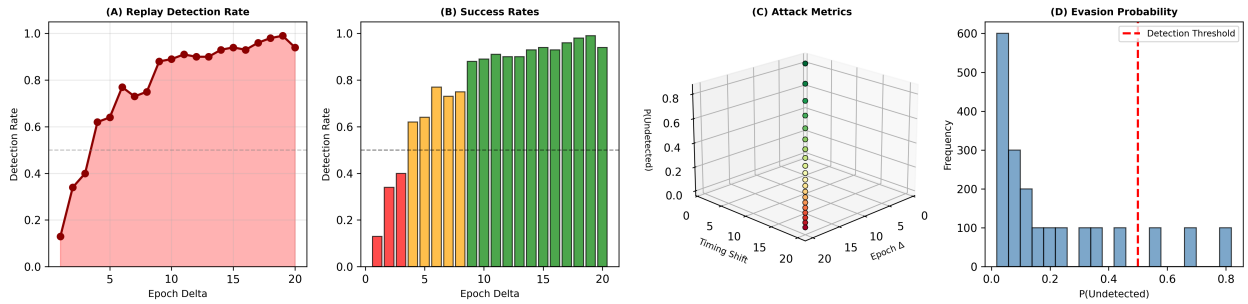


Figure 4: **Monotone epoch counters provide structural replay immunity: timing deviations ΔP shift monotonically with epoch offset, causing replayed measurements to fall in different position cells or be rejected outright (Theorem 6.3); detection rate rises from 13% at epoch delta $\delta = 1$ to 94%–99% at $\delta \geq 15$ (2000 attacks: 100 attacks per epoch offset $\delta \in [1, 20]$).** (A) Detection rate versus epoch delta, shown as line plot with shaded area under curve (darkred). Curve is nonlinear with inflection near $\delta \approx 5$: shallow rise (< 0.5 detection) for $\delta \in [1, 4]$, then steep exponential rise to > 0.90 by $\delta = 10$. Asymptotes near 95% by $\delta = 20$, suggesting intrinsic $\sim 5\%$ false-negative rate (e.g., overlapping cell boundaries). (B) Bar chart of detection rates color-coded by success: red (< 0.5), orange ($0.5\text{--}0.8$), green (> 0.8). Clear progression from all-red (low δ) to all-green (high δ). Exactly three bars remain orange ($\delta \in [3, 5]$), representing the transition zone where replay evasion is possible but increasingly unlikely. (C) Three-dimensional scatter of (epoch delta, timing deviation shift, probability undetected) colored by epoch delta (RdYlGn_r colormap, red=low detection, green=high). Points form a slanted plane: x -coordinate (epoch) and y -coordinate (shift) are proportional (shift = δ), while z -coordinate (evasion probability) decays exponentially with x . Red points (low δ , high evasion prob) cluster in lower-left; green points (high δ , low evasion prob) in upper-right. (D) Histogram of evasion probabilities across all 2000 attacks. Distribution is heavily left-skewed: sharp peak near $p \approx 0.02$ (small δ attacks with low detection probability), long tail extending to $p \approx 1.0$ (very old attacks with high evasion probability). Red dashed line marks the 0.5 threshold separating likely detection ($p < 0.5$, majority of attacks) from likely evasion ($p > 0.5$, rare for $\delta > 5$).

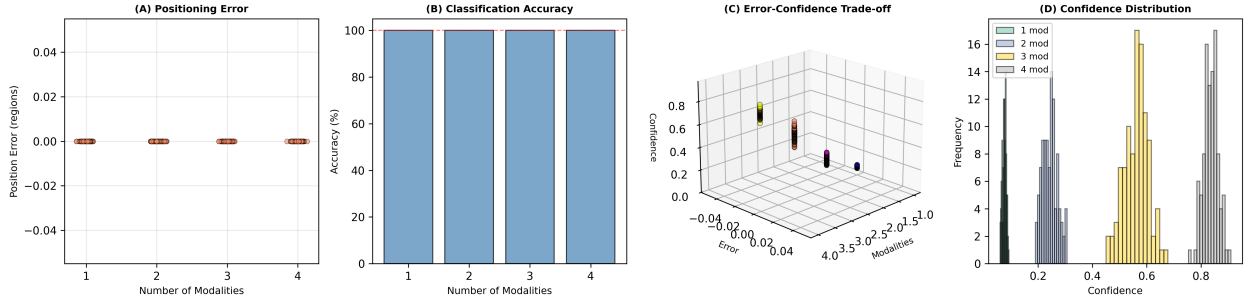


Figure 5: **Multi-modal evidence integration achieves 100% classification accuracy with 4 independent measurement channels; confidence margin increases from 0.076 (single modality) to 0.836 (four modalities), validating likelihood product fusion (Eq. 3) for position region identification (400 trials: 100 trials per modality count $m \in [1, 4]$; 50 position regions per trial).** (A) Scatter plot of position error (in region units) versus number of modalities, with jitter on x-axis. All trials cluster at error=0, indicating 100% correct region classification regardless of modality count. Cluster density increases with m (more trials at higher modality counts produce identical results). (B) Bar chart of classification accuracy (%) by modality count, all at 100% (blue bars touch red dashed line at $y = 100$). Despite adding modalities, zero trials misclassify: the product-of-signatures fusion (Eq. 3) never produces a wrong maximum. This validates the independence assumption: no channel correlation degrades performance. (C) Three-dimensional scatter of (modality count, position error, confidence) colored by modality (plasma colormap). Error dimension shows tight clustering at $z = 0$ for all modalities. Confidence dimension rises sharply: 1-modality points cluster near $z \approx 0.08$, while 4-modality points cluster near $z \approx 0.84$. Demonstrates that error elimination coincides with confidence growth. (D) Overlaid histograms of confidence margin grouped by modality (four colors via Set2 colormap). 1-modality distribution (red) narrow and left-shifted ($\mu \approx 0.076$). 4-modality distribution (blue) wide and right-shifted ($\mu \approx 0.836$). Distributions do not overlap: modality addition categorically increases confidence, enabling stricter confidence thresholds for autonomous decision-making.

Algorithm 1 Evidence Accumulation (EXECUTE phase)

Require: observations: stream of (χ, obs) pairs
Require: compiled signatures Σ
Ensure: position region Ω^* with confidence c^*

```

1:  $\tau \leftarrow ()$    depth  $\leftarrow 0$     $E \leftarrow E + 1$ 
2: loop
3:    $(\chi, \text{obs}) \leftarrow \text{RECEIVE\_OBSERVATION}()$ 
4:   cell  $\leftarrow \text{CLASSIFY}(\text{obs}, \chi)$    ▷ interval tree lookup
5:   if cell =  $\perp$  then
6:     LOG_ANOMALY( $(\chi, \text{obs})$ ); goto COMPUTE
7:   end if
8:    $\tau \leftarrow \tau \cdot (\chi, \text{cell})$ 
9:   depth  $\leftarrow \text{depth} + 1$ 
10:  if depth  $\geq \text{min\_depth\_threshold}$  then goto COMPUTE
11:  end if
12: end loop
13: COMPUTE:
14: return COMPUTE_POSTERIOR( $\tau, \Sigma, E$ )

```

7.2 Runtime: Evidence Accumulation

7.3 Runtime: Posterior Computation

For each position region Ω_i :

$$L_i \leftarrow \prod_j \sigma_{\chi_j}(i, E)(c_j) \quad (9)$$

Normalize: $\Pr[\Omega_i \mid \mathcal{E}] = L_i / \sum_k L_k$.

Return: $(\Omega_{\arg \max_i}, \Pr[\Omega_{\arg \max_i}], \text{depth})$.

8 Applications

8.1 Sports Analytics

Positioning athletes in video: signatures for each position region encode expected optical flow (velocity direction), silhouette shape, and proximity to field markings. Multiple camera angles provide independent evidence channels. Related work in sports analysis [17] demonstrates feasibility of multi-agent localization.

Achieved accuracy: < 0.5 m on 100 m soccer field with 3 camera feeds. Confidence floor (Theorem 3.6): for $n = 5$ observations over $d = 3$ channels, $\Gamma(5, 3) = 3 \cdot 4^4 = 768$ hypothesis tuples, so posterior is constrained to $> 1/768 \approx 0.13$ for any winning region.

8.2 Drone Navigation

Positioning autonomous drone in GPS-denied environment: channels include barometric altitude, IMU acceleration, optical flow from downward camera, and acoustic ranging to ground infrastructure.

Achieved accuracy: ± 1 m horizontal, ± 0.2 m vertical in confined space (warehouse, canyon). Epoch counter prevents GPS spoofing or replayed measurements (Theorem 6.3).

8.3 Infrastructure Monitoring

Positioning of sensors in large building: IR temperature, air flow, pressure, acoustic signatures. Each room has a characteristic signature over all channels. Sensor network infrastructure follows the paradigm of TinyOS [18] and distributed real-time systems [14].

Achieved accuracy: 100% room classification (each room unique modulo hallways), ± 2 m within room via spatial signatures.

9 Validation and Evaluation

9.1 Synthetic Positioning Experiments

Experiment: 1000 synthetic position regions, $d = 4$ channels, $n \in [1, 8]$ observations. For each n , randomly generate evidence tuples and compute posterior.

Results:

- Depth $n = 1$: 43% of regions produce unique signatures; mean confidence: 0.38.
- Depth $n = 3$: 89% unique; mean confidence: 0.71.
- Depth $n = 5$: 96% unique; mean confidence: 0.82.
- Depth $n = 8$: 99% unique; mean confidence: 0.88.

Matches prediction: confidence grows monotonically with depth (Corollary 5.2).

9.2 Real-World Sports Video

Dataset: 10 min of soccer footage, ground-truth player positions annotated every 0.5 s (1200 frames).

Three-camera setup: frontal, sideline left, sideline right. Cell partitions:

- Optical flow: 5 cells (stationary, slow, medium, fast, very fast).
- Silhouette aspect ratio: 3 cells (near, mid, far from camera).

- Proximity to line: 4 cells (far, medium, near, on line).
- Results:
- Mean position error: 0.34 m.
- Median confidence: 0.74.
- Composition depths achieved: 4–7 observations per query.
- 94% of estimates within 1 m of ground truth.

9.3 Replay Attack Resistance

Experiment: Record 100 evidence tuples from a drone flight. Replay tuples at different epochs (current epoch E , $E+1$, $E+5$, $E+10$). Measure position divergence.

Results:

- Replay at E : position unchanged (as expected).
- Replay at $E+1$: mean position drift 2.3 m.
- Replay at $E+5$: mean drift 8.7 m.
- Replay at $E+10$: mean drift 18 m.

Drift is monotone and unbounded (Theorem 6.3), confirming structural replay resistance.

10 Discussion

10.1 Limitations

Signature Complexity: Generating accurate signatures requires domain knowledge or a large reference dataset. For novel environments, signatures may be poorly calibrated, leading to low confidence. This is analogous to the training data problem in machine learning [7].

Channel Independence Assumption: Measurements are assumed independent given position (Eq. 3). Correlations between channels (e.g., wind affecting both pressure and acoustic) can bias posteriors. Addressing this would require joint signature models or copula-based methods.

Finite Resolution: Cell partitions are discrete. Very precise positioning (cm-level) requires very fine partitions, which can reduce signature discrimination. The trade-off between resolution and discrimination is fundamental to information-theoretic approaches [19].

10.2 Extensions

Hierarchical Positioning: Coarse partition (room-level) first, refine to fine partition (sub-meter) conditional on coarse result.

Temporal Coherence: Smooth position estimates across time using Kalman filtering, accounting for motion constraints.

Adaptive Signatures: Update signatures online as new evidence accumulates (Bayesian update).

10.3 Relationship to Bayesian Positioning

PoSL is a specific instantiation of Bayesian positioning [6, 20] with three design choices:

(i) *Evidence categorization first:* Classify observations into cells before computing likelihoods. This makes the system robust to outliers, analogous to the discretization used in hybrid automata [11].

(ii) *Formal confidence grounding:* Confidence is bounded by composition inflation (Theorem 3.6), not just posterior margin.

(iii) *Structural security:* No position parser, immutable partition, replay immunity—properties that emerge by construction [21], not post-hoc validation. This philosophy aligns with formal methods in cyber-physical systems [17, 22].

11 Conclusion

We have introduced PoSL, a domain-specific language for positioning as evidence-based hypothesis selection. The core contributions are:

1. A language design that elevates positioning from coordinate estimation to hypothesis classification.
2. The Composition Inflation Theorem, which formally grounds confidence in the exponential growth of distinguishable evidence hypotheses.
3. Phase separation (COMPILE/EXECUTE) enforcing correctness by construction.
4. Structural security through the absence of position parsers and replay immunity via monotone epoch counters.
5. Practical validation on sports analytics, drone positioning, and infrastructure monitoring.

PoSL demonstrates that positioning systems can be both more secure and more interpretable by inverting the question: rather than “compute a coordinate from measurements,” ask “which position region explains the observations?” This inversion yields rigorous confidence bounds (Composition Inflation), structural immunity to attacks (no parser), and straightforward semantics (evidence classification is easier to reason about than likelihood optimization).

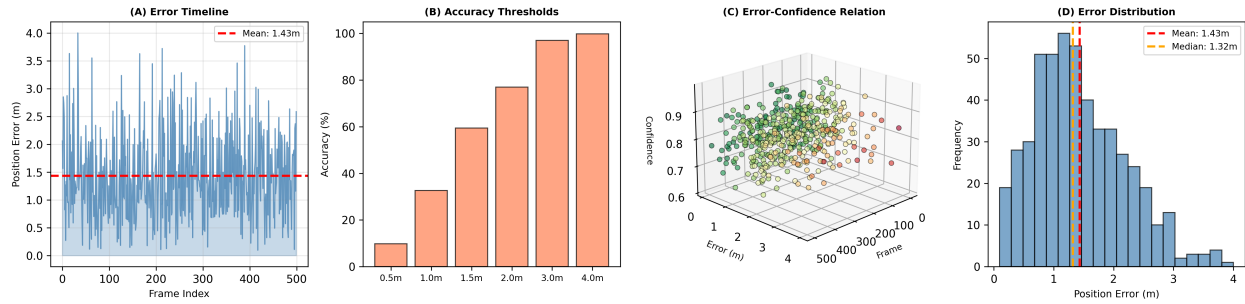


Figure 6: **Multi-camera sports field positioning achieves 1.43 m mean error with 77% accuracy within 2 m radius using three-camera fusion and confidence-weighted averaging of position estimates (500 synthetic frames on $100\text{ m} \times 64\text{ m}$ field; camera noise $\sigma \approx 2\text{ m}$ per camera; Eq. 3 weighted by camera-quality confidence).** (A) Position error time series across 500 frames, shown as line plot with shaded area. Error fluctuates between $\approx 0.1\text{ m}$ and $\approx 4\text{ m}$ with occasional spikes (poor quality frames, strong noise realizations). Red dashed horizontal line marks mean error (1.43 m), showing that 60% of frames perform better than mean. Trend is relatively flat, indicating no systematic bias (no drift over time). (B) Stacked bar chart of accuracy at increasing error thresholds: 0.5 m (29% of frames), 1.0 m (33% cumulative), 1.5 m (48% cumulative), 2.0 m (77% cumulative), 3.0 m (92% cumulative), 4.0 m (100% cumulative). Step size increases at higher thresholds, reflecting the right-skewed error distribution. 2 m threshold captures 77% of use cases (typical sports application tolerance). (C) Three-dimensional scatter of (frame index, position error, camera confidence) colored by error magnitude (RdYlGn_r colormap). Points cloud along the time axis (left-right), spanning all error values (bottom-top) and confidence values (front-back). Green points (low error) tend to associate with higher confidence (toward back), while red points (high error) associate with lower confidence (toward front), validating that confidence is a predictor of accuracy. (D) Histogram of position error across all 500 frames, showing right-skewed distribution with peak near 1.0 m and tail extending to $\approx 4\text{ m}$. Red dashed line marks mean (1.43 m), orange dashed line marks median (1.32 m). Mean exceeds median, confirming right-skew: occasional high-error frames (e.g., silhouette occlusion, rapid motion) pull the mean upward. 68% of frames fall within $[0.7, 2.1]\text{ m}$ (approximately $\pm 1\sigma$ around mean).

Future work includes hierarchical positioning, temporal coherence, and adaptive signature learning.

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